

# HITHA MAGADI VIJAYANAND

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## EDUCATION

### Texas A&M University

Aug 2024 – May 2026

Master of Computer Science - GPA : 3.77/4.00

### JSS Science And Technology University

Aug 2017 – Jul 2021

Bachelor of Engineering Computer Science - GPA : 9.52/10.00

## TECHNICAL SKILLS

**Language & Frameworks :** Java, Python, C, C++, JavaScript, Spring Boot, PyTorch, Ruby, Node.js, Typescript, HTML

**Tools & Platforms :** Git, Github, AWS, Kubernetes, Docker, Linux, Unix, Elasticsearch, Terraform, Jenkins, Jira, ServiceNow

**Databases & Others :** MongoDB, SQL, NoSQL, Postgres, DynamoDB, S3, Machine Learning, AI, PyTorch, TensorFlow, SDLC, Agile, Data Structures and Algorithms, REST APIs, Operating Systems, CI/ CD, DevOps, Containerization, Redis, Compliance, Statistics, Big Data Technologies, Infrastructure-as-Code (IaC), Cursor, Distributed Systems, Unit And Integration Testing, Data Mining

## PROFESSIONAL EXPERIENCE

### JP Morgan Chase

Jan 2024 – Aug 2024

Software Development Engineer 2

- Migrated microservices from a private to public cloud **AWS ECS**, **improving scalability** and **reducing deployment time** by 70%
- Redesigned the user notification workflow using an **event-driven architecture** powered by **RabbitMQ**, enabling the platform to handle and process over **15 million events** per month without performance degradation
- Developed a **real-time synonym training** system that fine-tuned NLP search models with evolving financial terminology, improving search accuracy by 30%
- Integrated **AI-assisted development** workflows using GitHub Copilot to improve developer productivity

### JP Morgan Chase

Aug 2021 – Dec 2023

Software Development Engineer 1

- Led the end-to-end modernization of the platform by upgrading **Spring Boot** from **2.0 to 3.0** and **Java 8** to **Java 17**
- Migrated a legacy MarkLogic system to a cloud-native architecture using AWS S3 and DynamoDB, improving scalability, resolving versioning issues and **reducing infrastructure costs by 80%**
- Designed **ETL data pipelines** to ingest, transform, and **index 1M+ records** into **AWS Elasticsearch**
- Created **interactive dashboards** for 5+ teams, streamlining data access and enhancing operational transparency

### JP Morgan Chase

Feb 2021 – Jul 2021

Backend Development Intern

- Automated business report generation via Outlook and Control-M, ensuring timely delivery and reducing manual effort
- Achieved 100% method coverage with **JUnit** tests and validated performance under load using **JMeter**
- Built scalable **REST APIs** for creating, updating, and managing **Elasticsearch indices**, streamlining data indexing workflows and enabling efficient configuration management and system administration

## PROJECTS

### Hybrid Self-RAG - Hallucination Reduction In LLMs | NLP, RAG, OpenAI

Jan 2025 – May 2025

- Built a **Hybrid Self-RAG** system using **Wikipedia** and **ArXiv APIs** improving factual grounding and accuracy by 9%
- Constructed a robust **FactScore-Based** Validation framework utilizing **Sentence-Transformer** embeddings and **cosine similarity** metrics to enhance the factual reliability of LLM outputs, increasing the average FactScore from 0.65 to 0.87

### Generator Matrix Optimization | Evolutionary Algo, Mathematics

Jan 2025 – May 2025

- Implemented a hybrid optimization pipeline combining **linear programming** with **genetic and simulated annealing algorithms** to minimize m-height in generator matrices for linear error-correcting codes
- Enhanced encoding efficiency and fault tolerance for high-reliability communication and storage systems

### Multimodal Network - Advanced Calorie Prediction | CNN, RNN, LSTM

Aug 2024 – Dec 2024

- Designed a multimodal deep learning architecture that integrates **Bi-LSTM** and **CNN** components to fuse CGM time-series data, neural signal features, and meal images, enabling more accurate and context-aware calorie prediction
- Achieved **0.33 RMSRE**, outperforming baselines and validating multimodal integration in health ML models

## CERTIFICATIONS AND ACHIEVEMENTS

- **Won** JP Morgan's **national-level hackathon** by developing a scalable solution to **transfer large media files** to **AWS S3** in Aug 2022
- **AWS Cloud Practitioner** certification issued by Amazon Web Services in Jul 2022